WEEK 2 · DAY 3, PART 1 · OVERVIEW

pandas

Combining and Loading, Part 1



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ABOUT 50 MIN WITH EXERCISES / 6 SECTIONS / 5 EXERCISES

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The Desk Lead's Twelfth Note

Today's data comes from three systems: positions, issuer details, and the rating scale. Put a name, a sector, and a rating on every position, and show me none was lost or doubled. Which benchmark bonds do we hold, and what weight do they carry? Then the book by sector, in money, beside the benchmark.

ABOUT 50 MIN WITH EXERCISES / 6 SECTIONS / 5 EXERCISES

Your Route Through Day 3, Part 1

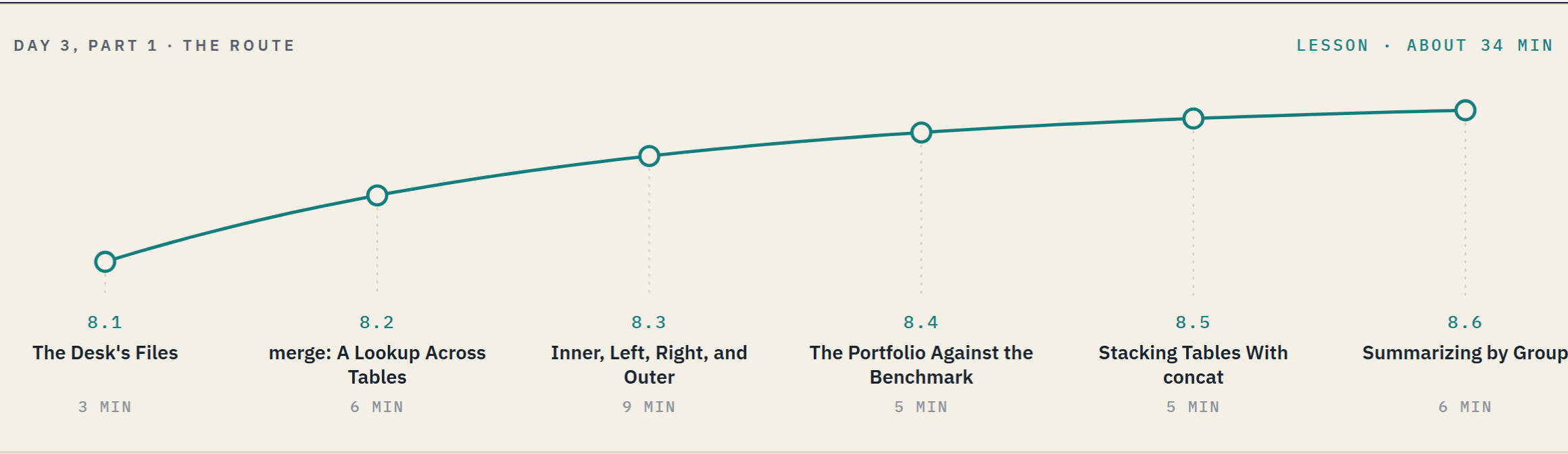


FIGURE 2 Six sections, in order, with about how long each takes.

Time. About 34 minutes for the lesson, then about 15 for the five exercises.

8.1 The Desk's Files

POSITIONS

	CUSIP	TICKER
0	99080CAE8	BEZE
1	99049XAE2	BEZH
2	99039NAC0	BEZL

ISSUERS

	TICKER	ISSUER
0	QVNE	Quorvane Energy
1	DNMR	Dunmarrow Rail
2	PLWK	Pellwick Paper

RATINGS

	RATING	SCORE
0	AAA	1
1	AA+	2
2	AA	3

MANY POSITIONS, ONE ISSUER

```
print('Tickers held:', positions['ticker'].nunique())
```

```
Tickers held: 115
```

Many rows on one side, one on the other: the shape of a lookup.

8.2 merge: A Lookup Across Tables

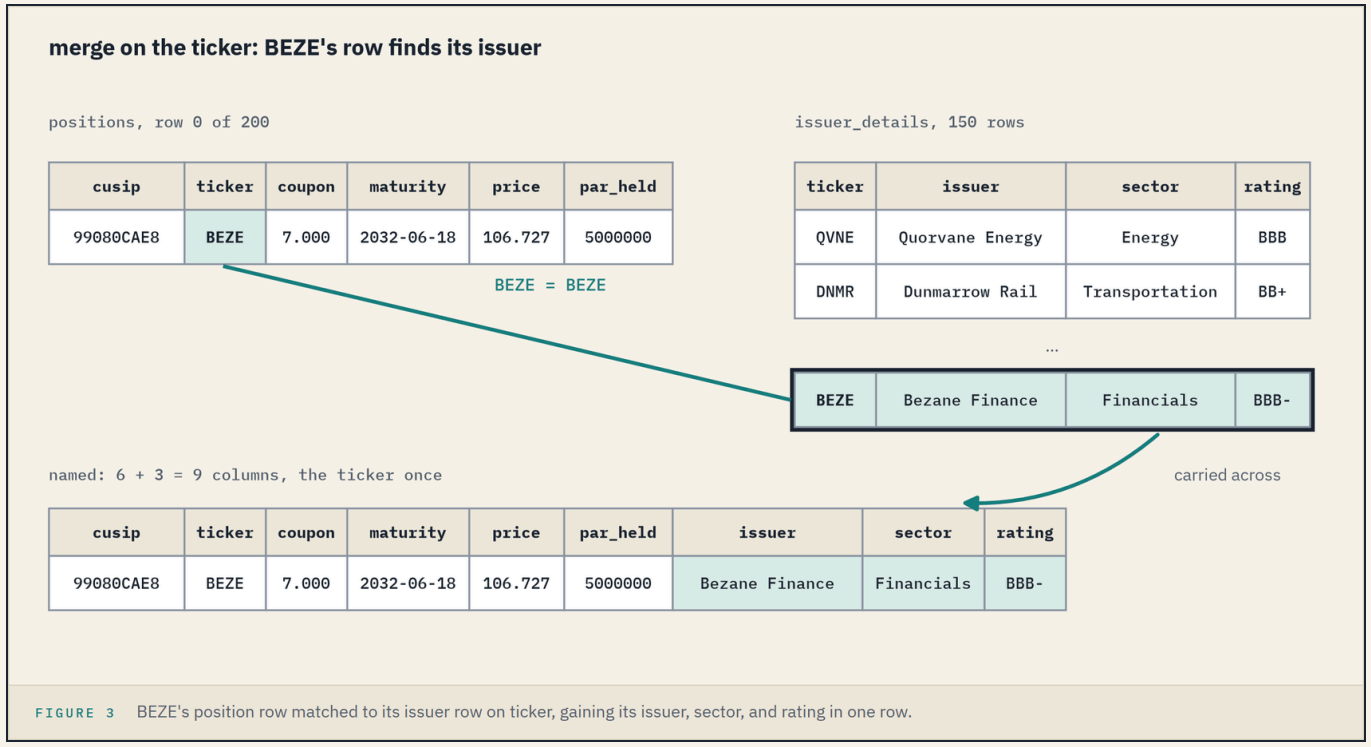


FIGURE 3 BEZE's position row matched to its issuer row on ticker, gaining its issuer, sector, and rating in one row.

```
named = pd.merge(positions, issuer_details, on='ticker')
print(positions.shape, issuer_details.shape, named.shape)
```

```
(200, 6) (150, 4) (200, 9)
```

EXCEL SIDE BY SIDE

JOB	PANDAS	EXCEL
Bring in the sector	`pd.merge(positions, issuers[['ticker', 'sector']], on='ticker')`	`=VLOOKUP(B2, issuers!A:C, 3, FALSE)` filled down

Three tables answered what one file could not: 128 investment grade, 72 high yield.

8.3 Inner, Left, Right, and Outer

LEFT: KEEP EVERY POSITION

```
left_join = pd.merge(desk_small, sector_small, on='ticker', how='left')
```

```
ticker     face          sector
0   QVNE  1500000          Energy
1   DNMR   500000  Transportation
2   PLWK   750000       Materials
3   HLVF   250000             NaN
```

OUTER: KEEP EVERYTHING, AND SHOW WHERE EACH WAS FOUND

```
pd.merge(desk_small, sector_small, on='ticker', how='outer', indicator=True)
```

```
ticker       face          sector      _merge
0   DNMR   500000.0  Transportation        both
1   HLVF   250000.0             NaN   left_only
2   PLWK   750000.0       Materials        both
3   QVNE  1500000.0          Energy        both
4   THAR        NaN       Materials  right_only
```

A Key Listed Twice

REFUSE A REPEATED KEY

```
pd.merge(desk_small, sector_twice, on='ticker', how='left', validate='many_to_one')
```

```
MergeError: Merge keys are not unique in right dataset; not a many-to-one merge

Duplicates in right:
 ticker
  QVNE ...
```

EXCEL SIDE BY SIDE

JOB	PANDAS	EXCEL
A key with no match	`NaN`, found with `isna()`	`#N/A`, found with `=ISNA(...)`
A key listed twice	rows repeated; `validate=` refuses	`VLOOKUP` takes the first match; `=COUNTIF(A:A, A2)` above 1 finds repeats

`validate=` refuses a repeated key; `isna()` and the `_merge` counts find a missing one.

8.4 The Portfolio Against the Benchmark

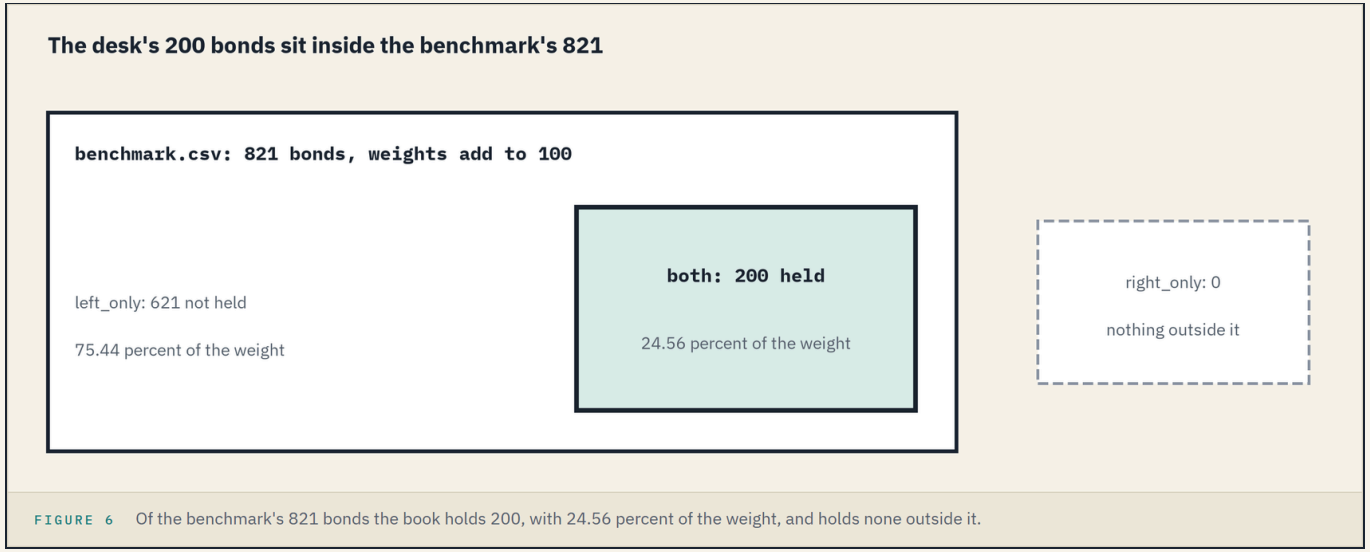


FIGURE 6 Of the benchmark's 821 bonds the book holds 200, with 24.56 percent of the weight, and holds none outside it.

HELD AND NOT HELD

```
print('Held:', held.sum(), 'Not held:', not_held.sum())
```

```
Held: 200 Not held: 621
```

EXCEL SIDE BY SIDE

On the benchmark sheet, a helper column T with `=COUNTIF(holdings!B:B, B2)` gives 1 for a bond held and 0 for one not; `=SUMIFS(S:S, T:T, 1)` adds the weights held, column S.

200 of 821 bonds, 24.56 percent of the weight, nothing outside it.

8.5 Stacking Tables With concat

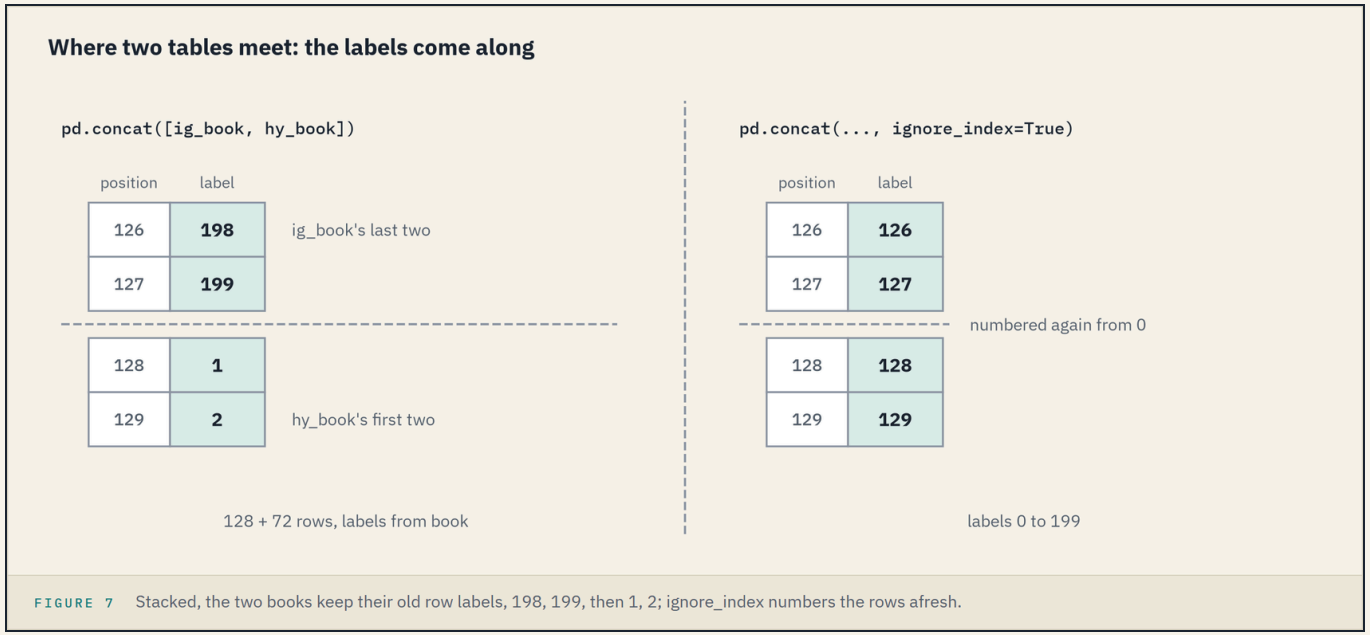


FIGURE 7 Stacked, the two books keep their old row labels, 198, 199, then 1, 2; ignore_index numbers the rows afresh.

STACK TWO TABLES

```
stacked = pd.concat([ig_book, hy_book])
print(ig_book.shape, hy_book.shape, stacked.shape)
```

```
(128, 22) (72, 22) (200, 22)
```

LABELS AT THE SEAM

```
print(stacked.index[126:130].tolist())
print(stacked.index[126:130].tolist())
```

```
[198, 199, 1, 2]
```

A key column: `merge`. Row labels, or one table under another: `concat`.

8.6 Summarizing by Group

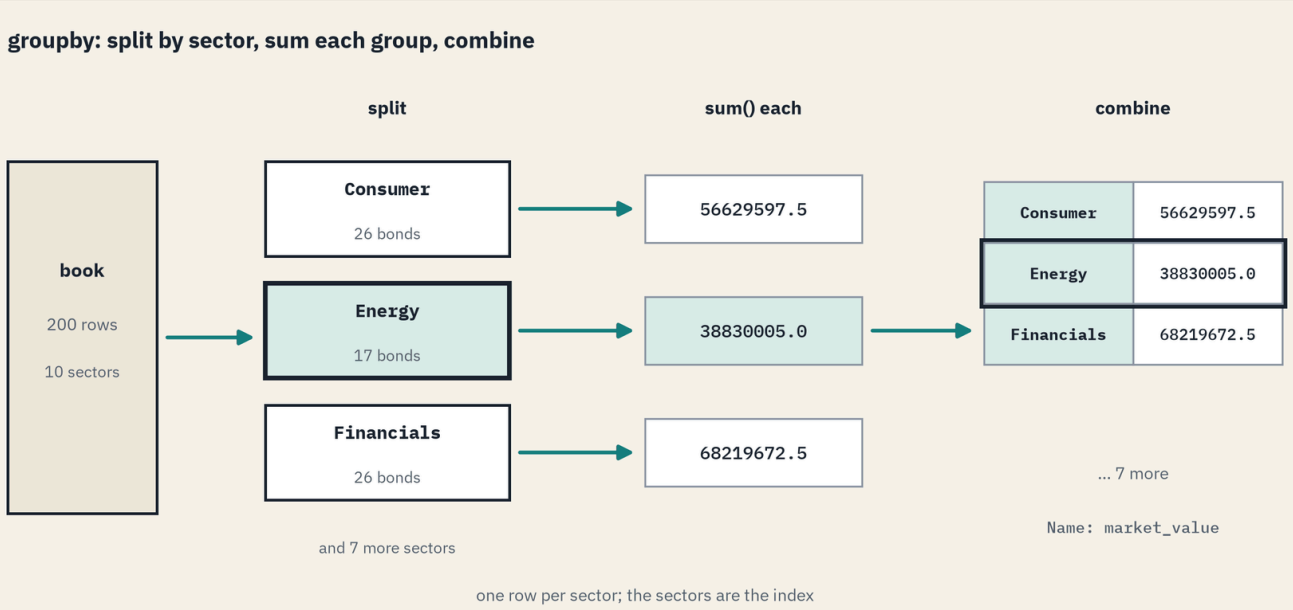


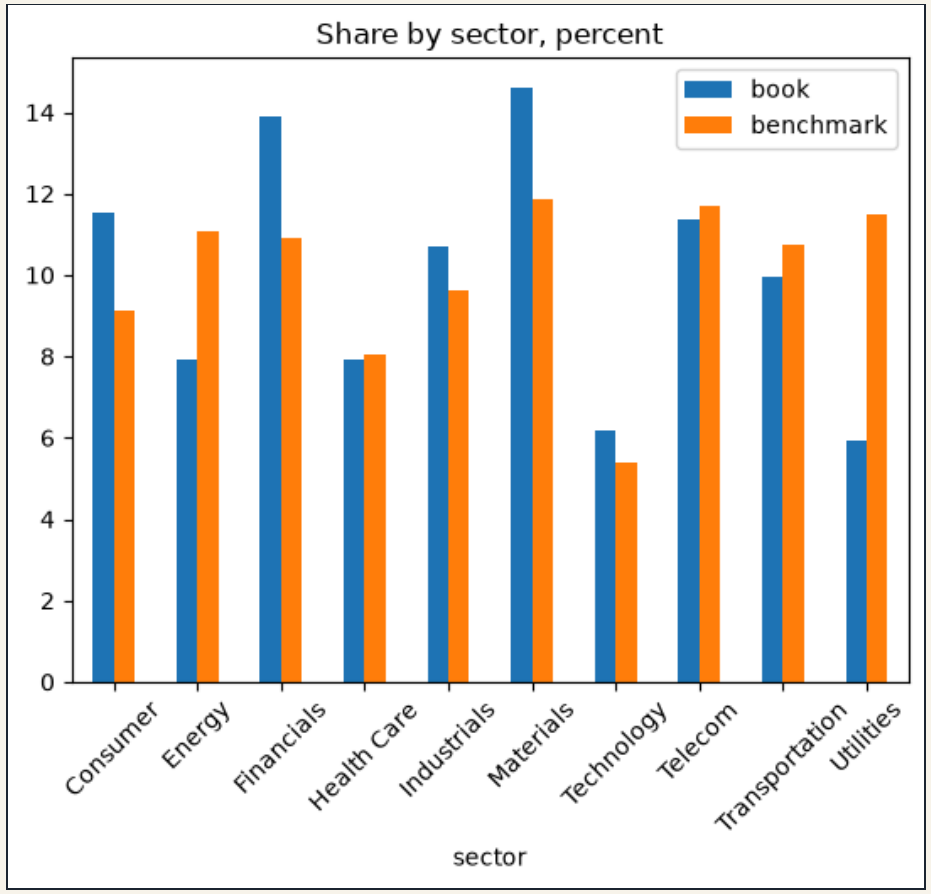
FIGURE 8 groupby splits the book into sectors, sums each group's market value, and combines the sums into one Series.

WHAT IT PRINTS

```
sector
Consumer          56629597.5
Energy            38830005.0
Financials        68219672.5
Health Care       38908187.5
Industrials       52540745.0
Materials         71701617.5
Technology        30311760.0
Telecom           55848270.0
Transportation    48822897.5
Utilities         29006462.5
Name: market_value, dtype: float64
```

One line does the last notebook's filter for every sector.

The Book Against the Benchmark



THE BOOK'S SHARE

```
book_pct = sector_value / sector_value.sum() * 100
```

THE BENCHMARK'S WEIGHTS

```
bench_pct = benchmark.groupby('sector')['weight_pct'].sum()
```

SEVERAL STATISTICS PER GRADE

GRADE	COUNT	MEAN	MEDIAN	MIN	MAX
High Yield	72	339.38	250.5	125	1551
Investment Grade	128	99.34	95.0	19	204

One `groupby` per question: 200 rows became ten sectors beside the benchmark.

You Can Now

- Join positions to issuer details and a rating scale with **`merge`**, as **`VLOOKUP`** does, and check the row count
- Choose **`inner`**, **`left`**, **`right`**, or **`outer`**, and find what did not match with **`indicator=True`** and **`isna()`**
- Guard a merge with **`validate='many_to_one'`** and an **`assert`**
- Compare a portfolio with its benchmark by CUSIP: bonds held and the weight they carry
- Stack tables with **`concat`**, and set them side by side with **`axis=1`**
- Summarize with **`groupby`** and **`agg`**, and set the book's share by sector beside the benchmark's

NEXT
WEEK 2 · DAY 3, PART 2

pandas: Combining and Loading, Part 2

Weighted averages, pivot tables, dates, and saving results to CSV files and Excel workbooks.